

PROJECTILE MOTION CALCULATOR

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ACADEMIC YEAR: 2026-2027

1.INTRODUCTION

This projectile motion calculator is a small Python console application designed to calculate common qualities related to motion of projectile. Instead of performing each formula manually. The user can select the required calculation from a menu, enter the initial velocity and angle and immediately obtain the result.

The program focuses on three quantities: time of flight, maximum height and horizontal range. It is intentionally kept simple so that the working of input, condition , functions and repetition can be understood clearly.

2. PROBLEM STATEMENT

Projectile motion formulas are easy to mix up when calculations are done manually, particularly when the same type of data of calculation has to be repeated for different values of velocity and angle . A small menu based program can reduce this repeated work.

This program accepts the value of initial velocity and projection angle and performs the selected projectile motion calculation . after displaying the the answer the program offers the user the choice to perform another calculation or end the program run.

3. OBJECTIVES

- Provide a clear number menu for the calculations
- Accept the initial velocity and angle values from the user
- Calculate time of flight , maximum of flight or range according to the selected option
- Display the calculated value in the console
- Allow another calculations during the run

4. FUNCTIONAL REQUIREMENTS

- Display the list of available projectile-motion calculation
- Read the user's menu choice
- Read the velocity of the projection
- Read the angle of projection
- Calculate the selected projectile motion parameter
- Print the result clearly
- Ask whether user wants to continue
- End program based on user choice

5. NON FUNCTION REQUIREMENTS

- Usability: the numbered menu and short prompts make the program straightforward
- Simplicity: The application uses basic python rather than unnecessary libraries
- Maintainability: Separate calculation sections/functions make the logic easier to modify
- Reliability: For the valid inputs the program should produce the corresponding calculated value
- Portability: Since the program uses basic python features it can be run of a system with Python 3.X installed

6. SYSTEM ARCHITECTURE

USER→MENU→OPTION SELECTION→INPUT VELOCITY AND ANGLE
→CORRESPONDING FUNCTION →CALCULATIONS→DISPLAY RESULT
→CONTINUE OR EXIT

7. PROCESS FLOW

START --> DISPLAY → WELCOME AND MENU → SELECT OPTION

(1-3) → ENTER VELOCITY AND ANGLE → CONVERT ANGLE TO RADIANS → CALL SELECTED FUNCTION → CALCULATE RESULT → DISPLAY RESULT → CONTINUE(Y/N) → IF YES REPEAT, ELSE STOP

8. DESIGN DECISION AND RATIONALE

The program defines one function per quantity ; time of flight, max height and range. Each takes velocity and angle of projection as parameters and converts the angle to radians for the calculation using the radians function of the math library. The calculation is carried out . The main program displays the menu, accepts the choice ,calls the corresponding functions prints the result and then asks if you want to continue. Gravity is takes as 10 m/s^2 .

9. TECHNOLOGIES USED

- Python 3.X
- Math library
- Functions
- If-else statements\
- While loop
- Basic arithmetic and physics formulas

10. SCREENSHOTS

- MAIN MENU

```
WELCOME
Please make a choice from the below listed options
1. Time of flight
2. Maximum height
3. Range of projectile motion
Enter the corresponding number of your preferred option:
```

- TIME OF FLIGHT

```
Enter the corresponding number of your preferred option:1
Enter the velocity of the object:20
Enter the angle of projection:30
The time of flight will be : 1.9999999999999996
Do you want to continue?(y/n)
```

- MAXIMUM HEIGHT

```
Enter the corresponding number of your preferred option:2
Enter the velocity of the object:20
Enter the angle of projection:30
The maximum height will be : 4.9999999999999998
Do you want to continue?(y/n)
```

- RANGE

```
Enter the corresponding number of your preferred option:3
Enter the velocity of the object:20
Enter the angle of projection:45
The range will be : 40.0
```

11. TESTING APPROACH

Test Case	Input	Expected Result
Time of flight	v=20, angle = 30	T=2.0s
Maximum height	v=20, angle =30	H=5.0m
Range	v=20 , angle =45	R=40m
Invalid	Choice=5	"invalid input"

12. CHALLENGES FACED

- Understanding functions creation and function calls
- Passing user inputs into calculations

- Converting angles from degree and radians
- Using the math library and its alias
- Connecting menu choices with functions
- Using a while loop for repeated calculations
- Handling invalid menu choices
- Debugging variable name errors and name conflicts , such as overwriting the math alias

13. FUTURE ENHANCEMENTS

- Add validation for negative or zero velocity and for angles outside 0-90 degrees
- Use math.pi and more precise $g=9.8$
- Accept decimal velocity
- Plot the projectile's trajectory with matplotlib
- Add initial-height and air resistance
- Show all three results in a single run
- Create a graphical user interface
- Improve error handling

14. REFERENCE

VITyarthi - Build Your Own Project: General Project Instructions & Submission Guidelines. Python Essentials course concepts and classroom material.

15. CONCLUSIONS

The Projectile Motion Calculator shows how basic Python concepts can be combined to build a useful physics tool. The project uses functions, the math library, user input, conditional statements, loops and arithmetic to calculate the time of flight, maximum height and range of a projectile. It is a simple example of modular, menu-driven programming.

